Life Cycle Cost Analysis of Pavement Design Alternatives for Collision-Free Roads

Yared Dinegdae orcid.org/0000-0001-7174-7214
Senior Researcher
Pavement Technology
VTI, Swedish National Road and Transport Research Institute, Linköping, Sweden, SE-581 95
Email: yared.dinegdae@vti.se

Abubeker Ahmed orcid.org/0000-0002-6327-4709
Senior Researcher
Pavement Technology
VTI, Swedish National Road and Transport Research Institute, Linköping, Sweden, SE-581 95
Email: abubeker.ahmed@vti.se

Jiqing Zhu orcid.org/0000-0003-1779-1710
Senior Researcher
Pavement Technology
VTI, Swedish National Road and Transport Research Institute, Linköping, Sweden, SE-581 95
Email: jiqing.zhu@vti.se

Sigurdur Erlingsson orcid.org/0000-0002-4256-3034
Professor
Pavement Technology
VTI, Swedish National Road and Transport Research Institute, Linköping, Sweden, SE-581 95 and
Faculty of Civil and Environmental Engineering, University of Iceland, Hjardarhagi 2-6, IS-107
Email: sigurdur.erlingsson@vti.se

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ABSTRACT

There have been efforts in Sweden to convert and transform single carriageway roads into collision-free roads. The structural design and cost-effectiveness of these roads have mainly been performed using empirically derived pavement design methods. Currently, the Swedish Transport Administration (STA) is financing a project aimed at developing and implementing a mechanistic-empirical (M-E) pavement design approach for Swedish conditions. This is expected to impact long-term pavement performance and subsequently, the total life cycle cost (LCC) of projects. This paper examines, through design alternatives, the impact of a pavement design method on the total LCC of a road project. This is achieved by designing and analyzing an actual flexible pavement structure using two empirical pavement design methods and one M-E pavement design method. A life cycle cost analysis (LCCA) was employed to estimate future maintenance and rehabilitation (M&R) and road user costs, as well as initial construction and end-of-life costs. The cost estimation was conducted considering various combinations of interest and inflation rates that are expected to reflect current and future cost trends. The analyses have shown that the M-E pavement design approach delivered the least LCC, but this LCC is observed to vary considerably depending on the inflation and interest rates employed.

Keywords: ERAPave PP, life cycle cost, collision-free, pavement design, inflation, interest rate

INTRODUCTION

The Swedish Transport Administration (STA), through its Vision Zero program, aims to significantly reduce the number of fatal accidents on Sweden's state roads. This initiative prompted extensive investment to improve and enhance traffic safety within the road network. One noticeable outcome of this effort is the introduction of collision-free roads. This road type introduces a cable barrier as a separation unit and alternates the overtaking possibility between the two directions, effectively improving traffic safety. In addition, when constructed on an existing single carriageway road, this road type is expected to require considerably lower investment, as it causes less land intrusion (*1, 2*). Collision-free roads, though considered socioeconomically profitable, have raised concerns within the STA regarding the cost-effectiveness of future operation and maintenance activities (*1*).

Collision-free roads are mainly constructed by widening certain sections of an existing single carriageway road, resulting in narrower lanes. The narrow lanes, combined with the cable barriers, have been observed to reduce the lateral wander of vehicles and create more channelized traffic (*3*). As observed in pavement condition surveys, the annual rate of rutting formation on collision-free roads is up to three times higher compared to single carriageway roads. This is expected to significantly increase future maintenance and rehabilitation costs (M&R) (*1*). Furthermore, the difficulty of performing M&R activities is expected to increase costs associated with winter maintenance, crash barriers and road users. The cost-effectiveness of collision-free roads has been the subject of many studies that combine empirical performance data with either cost-benefit analysis (CBA) or LCCA (*1, 4, 5*).

Service life is one of the inputs utilized in LCCA to determine the M&R costs of roads. It also plays an important role in estimating road user costs (RUC) (*6*). In Sweden, it is customary to use empirical pavement design methods to determine the long-term performance of flexible pavements (*4, 5*). These design methods were developed based on either traffic volume or inputs such as lane width and layer thickness. As pavement degradation is a complex process involving several factors and inputs, the use of empirical methods can result in erroneous service life predictions, subsequently affecting the accuracy of the estimated total LCC (*7*). Furthermore, factors such as climate change, technological developments, advancements in material performance and construction techniques will further limit the capability of empirical pavement design methods (*8*).

The STA is currently developing a novel M-E pavement design method for flexible pavements with the goal of addressing existing and future challenges. The method, Elastic Response Analysis of Pavements - Performance Predictions (ERAPave PP), optimizes flexible pavements for distresses such as cracking and rutting. It can also be used to evaluate distresses typical of cold climate regions, such as frost heave and studded tire wear (*9, 10*). The evaluation of these distresses is achieved using prediction models that consider the underlying pavement degradation mechanisms through the incorporation of advanced material models and pavement response modeling. Unlike empirical methods, ERAPave PP offers a detailed characterization of traffic and climate inputs, allowing predicted results to accurately capture the impact of these factors on long-term pavement performance (*11, 12*). The adoption of ERAPave PP is expected to improve not only pavement performance evaluation but also the economic and environmental impact assessment of road projects.

OBJECTIVE

The main objective of this research is to investigate and establish the impact of a pavement design method on the total estimated LCC of a road project. A traffic volume-based empirical method, a service life based empirical method, and ERAPave PP are the three investigated pavement design approaches. In addition, the study assesses the impact of various combinations of inflation and interest rates on the estimated LCC. This is achieved using an actual road section that was previously converted from a single carriage-way into a collision-free road. Future costs were estimated and compared using economic indicators such as net present value (NPV) and equivalent uniform annual cost (EUAC). A further breakdown of agency costs and RUCs was performed to identify cost components that drive the total LCC.

LIFE CYCLE COST ANALYSIS

Life Cycle Cost Analysis (LCCA) is a methodology for the systematic economic evaluation of a project over its lifetime (*13*). As it systematically accounts for current costs and the key factors that affect future costs, LCCA has been used in the paving industry to choose investment strategies that are economically efficient (*14*). It is highly recommended that LCCA be considered as early as possible to increase the potential for making financially sound decisions (*15*). Furthermore, LCCA should be implemented at all stages of the investment decision-making process to aid in selecting between alternative projects, construction techniques and building components. The common practice in the paving industry, however, is to use LCCA to compare two or more projects with equal or similar benefits (*16*).

A road LCCA involves a multitude of cost items with varying levels of impact on the total estimated cost. A simplification of the analysis by excluding cost items that are small, identical, and without any impact on future decisions, is recommended (*17*). It is suggested also to disregard RUC in normal usage when similar pavement designs are compared (*14*). It is recommended, however, to include road user and external costs along with agency costs for M&R treatments when there is a significant difference between alternatives (*18*). Based on the party that bears the burden, the cost items in road LCCA can be categorized into agency, user and societal costs (*19*). A representation of this cost-breakdown is presented in Figure 1.

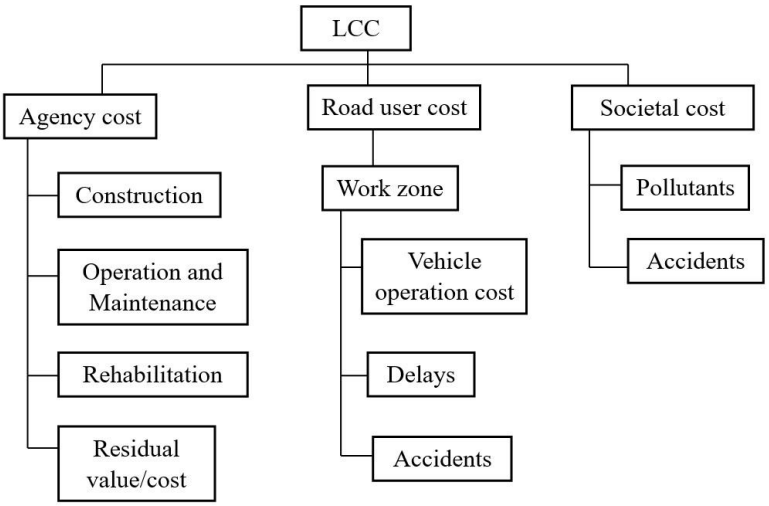


Figure 1 Cost breakdown of a road life cycle cost (LCC)

Although it is an important part of road LCCA, the residual or end-of-life value/cost of a road project is difficult to measure and quantify. It is common to use other indirect functional requirements to measure the residual value/cost of a road after the warranty period is completed (*20, 21*). The residual value/cost of a road is commonly considered to have two components. The first component refers to the net value that can be derived from pavement recycling, while the second component describes the remaining service life beyond the analysis period. In this study, a residual cost approach which is based on the principle that an asset should be restored to an acceptable condition at the end of the analysis period was employed. Accordingly, Equation 1 is used to estimate residual costs (RC) (*5, 16*). In Equation 1, *Age of treatment* represents the time from the last treatment to the end of the analysis period, *Service life* is the treatment's effective lifespan, and *Treatment cost* is the expense associated with the rehabilitation measure.

$$RC = \frac{Age\ of\ treatment}{Service\ life} \cdot Treatment\ cost \quad (1)$$

The NPV is one of the most commonly used economic indices in road LCCA. The index represents the monetary value that must be invested today to cover all expected costs over the analysis period (*22*). The basic equation to calculate the NPV of the LCCs is presented in Equation 2.

$$NPV = C + \sum_{k=1}^{N} Future\ Cost_k \left[\frac{1}{(1+r)^{n_k}}\right] + \frac{RC}{(1+r)^N} \quad (2)$$

where C is the initial construction cost, RC is residual cost, r is the discount rate and N is the analysis period.

EUAC is another commonly used economic index in LCCA. It represents the annual cost required to own an asset over its lifetime and is often used as a decision support tool to compare the cost-effectiveness of alternatives with varying lifespans (*22*). Equation 3 presents the mathematical formulation for EUAC.

$$EUAC = NPV \times \left[\frac{r\,(1+r)^N}{(1+r)^N - 1}\right] \quad (3)$$

In most LCCA studies, the cost items included are determined by the user, depending on the object under study. It is recommended to perform the analysis considering factors that have significant impact on the result (*13*). In this study, the analyses were performed considering both agency and road user costs. In addition, the impact of inflation and interest rate fluctuations on LCC was examined.

Discount rate

Discount rate, which roughly represents the difference between interest and inflation rates, indicates the real value of money over time (*23*). Inflation is commonly measured using the consumer price index (CPI). As CPI does not accurately reflect the cost fluctuation in the construction industry, the STA recommends a road construction cost index called 'Vägindex'. Although, both CPI and Vägindex are characterized by considerable fluctuations, the STA recommends a constant discount rate (*24*). Figure 2 presents the trend of interest and inflation rates in Sweden over the past 25 years. As shown in the figure, both rates exhibit significant fluctuations, which, if not properly accounted for, can lead to serious economic consequences.

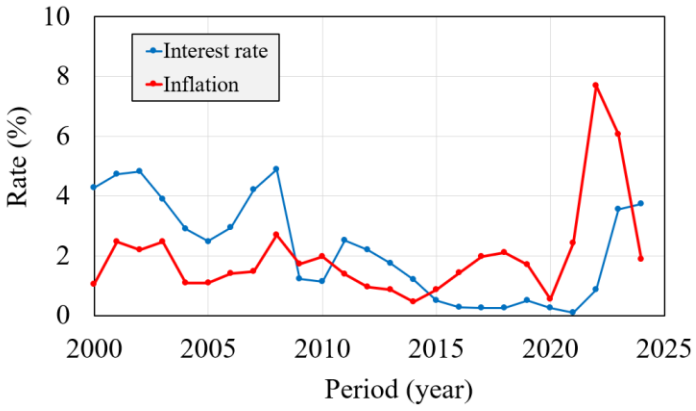


Figure 2 Swedish lending rates and consumer price index (CPI) inflation trends

For this study, a CPI inflation rate of 3% and a Vägindex rate of 5% were used to compare the impact of inflation on estimated LCC. These rates represent an average yearly increase over a 20-year

period (*25*). For estimating the NPV of the studied design alternatives, a discount rate of 3.5% was used (*5*).

Agency cost

Agency costs are expenses that are often covered by road authorities throughout the lifetime of a road project. These costs can be divided into parts such as design, construction, operation, M&R, and end-of-life (*26*). TABLE 1 presents the unit prices for the surface and binder layers for construction, along with the cost for the surface layer for M&R. These prices reflect the price levels for the year 2010 and were obtained by consulting various sources (*5*). All activities, including the construction of the surface course, require milling and traffic management equipment, which cost 20 SEK/m^2 and 26 SEK/m^2, respectively. M&R activities occurring during the first 20 years involve only the milling and replacement of the surface layer. However, the first M&R activity after 20 years involves reconstruction of the binder layer, along with milling and placement of a new surface curse.

TABLE 1. Unit Prices for Agency Cost

Activity	Layer	Mixture	Thickness (mm)	Unit price (SEK/m^2)
Construction	Surface	ABS16 70/100	40	106
	Binder	ABb16 50/70	60	129
M&R	Surface	ABS 16 70/100	40	85

Road users cost

Road user cost (RUC) refers to the cost that is burdened by road users in the work-zone during construction, and M&R activities. RUC can generally be categorized into three main components: vehicle operation cost (VOC), time cost (delay) and accident cost (*27*). The RUCs of the evaluated design scenarios were achieved using the unit prices in TABLE 2. As agency costs, these prices reflect the price levels of year 2010. The RUCs for the work-zone were estimated using a detoured road in a unit cost per closed road (*5*).

TABLE 2. Unit Prices for RUC

Cost category	Unit price (SEK/m)
Delays	730
VOC	179
Accidents	70
External costs	17

DISTRESS MODELS AND PAVEMENT SECTION

The structural design of a pavement section is highly governed by the design method employed (*28*). Different design methods are expected to result in varying service lives, impacting estimated agency and road user costs. In this study, two empirical and one M-E pavement design methods were used to analyze and design the studied pavement structure. A performance criterion of 14 mm rut depth, which is a limiting requirement by the STA, was used to establish the service life for M&R interventions (*29*). Rutting was selected as a performance criterion as it is often the limiting distress mode that triggers pavement maintenance and rehabilitation activities.

Traffic-based empirical method

Collision-free roads deteriorate at a higher rate than single carriageways, which is mainly attributed to the reduction in vehicle lateral wander. Traffic flow measurements have shown that on single-lane roads, the reduction in the standard deviation of the vehicle lateral wander reaches almost 40% for both passenger

cars and heavy goods vehicles (HGVs) (*30*). This channelized traffic is expected to induce more studded tire wear and permanent deformation. In addition, the construction of collision-free roads on existing weakened structures can contribute to a higher rate of pavement deterioration (*4, 31*).

Traffic and performance data from the Swedish pavement management system (PMS) were used to establish the traffic-based pavement deterioration model. In the model, a one-way annual average daily traffic (AADT) was used to establish the annual rate of rut development (*4*). Equation 4 presents the regression equation that defines the relationship between AADT (10^3) and annual rutting (mm/year). For low-traffic volume sections, Equation 4 predicts longer service lives, and this service life decreases as the traffic volume rises. The incorporation of traffic volume allows the model to capture the increased rate of deterioration that accompanies the aging pavement structure as well as wear due to studded tires.

$$annual\ rutting\ (y)\ = 0.187 * AADT(y) + 1.058 \quad (4)$$

Service life-based empirical method

Rutting in flexible pavements arises due to a combination of both studded-tire wear and permanent deformation. The wear resistance of the surface course is mainly governed by factors such as maximum aggregate size, aggregate quality, and lateral traffic wander. On the other hand, the resistance to permanent deformation is governed by the overall structural response of the pavement structure to the prevailing environmental and traffic loads (*32, 33*). As it involves a complex degradation process with several inputs and factors, the prediction of rutting development in flexible pavements should be performed incorporating both structural and mixture inputs.

A service life-based empirical performance prediction model was developed considering inputs from both the surface layer and structure. The development of the model was achieved in two steps. In the first step, wear and permanent deformation predictions, using a regression-based abrasion model and a linear elastic multi-layered model, respectively, were validated using the Swedish LTPP database (*32, 33*). Afterwards, the validated models were used to predict the rutting performance of existing collision-free roads, facilitating the development of a linear relationship between service life and inputs such as lane width (l_w), aggregate quality (A_n) and binder layer thickness (t_h) (*5*). The Nordic abrasion value is used to determine the aggregate quality (A_n). Equation 5 presents the mathematical formulation for the model. Although it incorporates several inputs, the service life-based model, unlike the traffic-based model, does not capture the rate of deterioration associated with the aging pavement structure.

$$service\ life = -1.27 + 3.00 * l_w - 0.43 * A_n + 0.03 * t_h \quad (5)$$

ERAPave PP

ERAPave PP employs M-E principles to optimize flexible pavements for distresses such as rutting and fatigue cracking. The rutting or permanent deformation prediction models have been evaluated and calibrated using performance data from laboratory, accelerated pavement testing (APT) structures, as well as field sections, and have been shown to deliver acceptable results (*9, 34*). A comprehensive evaluation of the tool is currently underway, including the calibration and validation of the fatigue cracking model using field sections. The tool, which is under consideration by the Swedish and Norwegian Transport Administrations as a potential successor to existing methods, is expected to address many of the challenges currently facing the pavement industry.

The design procedure implemented in ERAPave PP is presented in Figure 3. As shown in the figure, the tool consists of an input module and a pavement analysis component. The input module provides all the required information for input characterization. A heat balance model is used to determine the temperature profile in the asphalt concrete (AC) layer and the frost depth in the substructure layers. Axle load spectra (ALS) are used to characterize truck traffic, while mechanical properties such as dynamic and resilient moduli are determined based on inputs such as temperature, loading frequency and moisture content (*11, 12*). An extensive network of weather stations from both countries is integrated into

the tool, allowing the extraction of accurate historical climate data. In addition, the tool includes a large database of information on asphalt mixtures, unbound granular materials and subgrade soils.

The pavement analysis component performs several evaluations focused on structural capacity. It performs bearing capacity and frost action controls to ensure structural integrity for extreme loading and frost heave damage, respectively. It also computes distresses based on empirical models that take pavement response outputs from a multi-layered linear elastic theory (MLET)-based analysis. It further conducts performance evaluation and selects an optimum structure by comparing accumulated distresses with the corresponding design criteria.

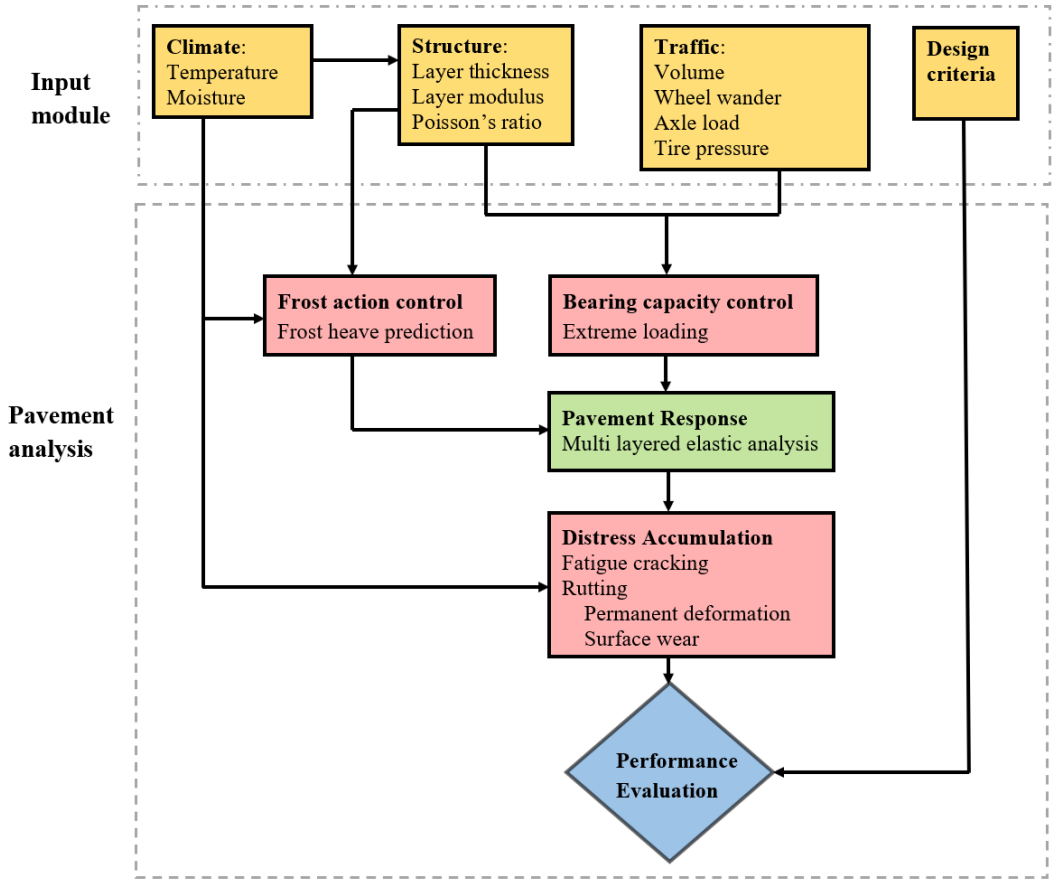


Figure 3 Design flow chart for ERAPave PP

Rutting is estimated in ERAPave PP considering both permanent deformation and surface wear due to studded tires. The tool uses the layer strain approach to predict the permanent deformation behavior of the pavement structure. In this approach, unique models or model coefficients are utilized to determine the contribution of each layer to the total permanent deformation (*9, 35*). Equations 6 and 7 present the permanent deformation prediction models used in ERAPave PP for the asphalt and unbound granular layers, respectively. As shown in Equation 6, permanent deformation (ε_P) is determined considering factors such as applied traffic cycles (N), temperature (T) and applied load level. For the unbound granular layers and subgrade, the inputs considered are applied traffic cycles (N) and resilient strain (ε_r) (*36, 37*).

$$\varepsilon_P\,(N,T) = aT^bN^c\varepsilon_r \qquad (6)$$

where ε_P and ε_r are permanent and resilient strains, respectively. a = 0.03, b = 1.85 and c = 0.27 are model parameters.

$$\varepsilon_P\,(N) = aN^{b(\varepsilon_r)}\varepsilon_r \qquad (7)$$

where a = 3.0 and b, which has values of 100, 200 and 250 for base, subbase and subgrade, respectively, are model coefficients.

The Swedish abrasion prediction model is incorporated into ERAPave PP to determine the surface wear due to studded tires. This model consists of two sub-models: the mineral aggregate sub-model and the Prall sub-model. Both models deliver acceptable results and were developed using performance data from a full-scale road simulator and experimental field sections (*33, 38*). For this study, the mineral aggregate sub-model was used, as required inputs such as the Nordic abrasion value (A_N), maximum aggregate size (D_{max}), and the mineral aggregate share with diameter > 4 mm (MAS_4) can easily be established. Equation 8 presents the mineral aggregate sub-model, where the annual wear rate along the profile ($W(x)$ mm/year) is calculated using inputs such as number of vehicles ($T(x)$), material factor (f_M) and the abrasion wear per vehicle (W_{veh}). W_{veh}, which has a value between a millionth of 10 mm and a millionth of 50 mm per vehicle, varies depending on the type of studded tire, wearing course mixture, and aggregate quality (*12, 38*).

$$W(x) = T(x) \cdot f_M \cdot W_{veh} \qquad (8)$$

T(x), which represents the volume of light traffic with studded tires, is determined using inputs such as annual average daily light traffic ($AADT_L$), the share of light vehicles with studded tires, and the length of the wear period. The distribution of this traffic along the transverse direction ($T(x)$) is achieved using the lateral wander of the traffic, as well as a normal distribution function. On the other hand, the material factor, which accounts for the impacts of the wearing course, stud type, operational condition, and traffic speed, is determined using Equation 9.

$$f_m = (2.493 + 0.144 \cdot A_n - 0.069 \cdot D_{max} - 0.017 \cdot MAS_4) \cdot f_v \cdot f_s \cdot f_{st} \qquad (9)$$

where f_v, f_s and f_{st} are factors that represent the average speed of the traffic, the presence or absence of de-icing, and the stud calibration, respectively.

Pavement section

The study was conducted using an actual pavement structure, Road RV35, located in Östergötland county, Sweden, as a case study. The road, which is a collision-free asphalt road, has a total length of 6.1 km and a lane width of 3.3 m. The structural profile consists of two bituminous layers: a 40 mm wearing course and a 60 mm binder course, and two unbound granular layers: 100 mm base course and a 460 mm subbase course (*4, 5*). The structure is constructed on a clay foundation with type 4e-Lera. For the construction of the wearing course, aggregates with a Nordic abrasion value of 5 were used. The mixture types for the wearing and binder courses are provided in TABLE 1.

The first-year traffic of the section, as measured by AADT, is 7800 and this traffic volume is expected to grow annually by 1.8%. The road is designed for a speed limit of 100 km/h and HGVs are expected to comprise 6.6% of the total traffic. Standard deviations of 204 mm and 178 mm for light vehicles and heavy-duty vehicles, respectively, were used to model the lateral wander of the traffic (*4, 5*). It is expected that 65% of passenger cars will have studded tires for the winter season. For modeling environmental factors, hourly climate data, including air temperature, were extracted from a local weather station, while moisture information for the unbound granular layers and subgrade was obtained from design guides.

RESULTS AND DISCUSSION

Long-term performance

The studied pavement structure was analyzed using the three design approaches to determine the required frequency of M&R interventions. A 40-year analysis period, which is commonly employed as an economic period for Swedish roads, was used for the evaluation (*39*) . Table 3 presents the intervention periods, starting from when the road becomes operational, along with the expected treatments and the corresponding NPV costs in SEK/m. As shown in TABLE 3, the service life between treatments get shorter for both the traffic-based method and ERAPave PP, while it remains the same for the service life-based method. Average service lives of 7, 8, and 9 years were obtained for the traffic-based method, the service life-based method and ERAPave PP, respectively. Although the three methods were developed based on different design assumptions and philosophies, the estimated average service lives are comparable.

TABLE 3. Comparision of Predicted Performance

Traffic-based method			Service life-based method			ERAPave PP		
Year	Treatment	NPV	Year	Treatment	NPV	Year	Treatment	NPV
8	Mill & replace	2,141	8	Mill & replace	2,141	10	Mill & Replace	2,120
15	Mill & replace	2,069	16	Mill & replace	2,059	19	Mill & Replace	2,030
22	Reconstruction	2,597	24	Reconstruction	2,572	27	Reconstruction	2,534
28	Mill & replace	1,943	32	Mill & replace	1,906	34	Mill & Replace	1,887
34	Mill & replace	1,887	40	Residual	555	40	Residual	476
40	Residual	555						

LCC estimation

The LCC of each design alternative was calculated using the M&R intervention frequencies provided in TABLE 3 and the unit prices listed in TABLE 1TABLE 2. For estimating the LCC, a functional unit defined by a road length of 1 m and a lane width of 3.3 m was used. Costs associated with the construction of unbound granular layers, winter maintenance and barrier construction were excluded from the analyses, since these costs are the same for all the three alternatives. Construction costs were calculated using 2025 as the base year, requiring the unit prices listed in Tables 1 and 2 to be adjusted using the designated inflation rate of 3%. Accordingly, a cost of 1208 SEK/m was estimated for the construction of the surface course and binder layer, which is the same for each design alternative. In addition, the M&R costs and RUCs for each intervention period were calculated by inflating the listed unit prices according to the time of the intervention. For calculating the residual costs at the end of the economic period, Equation 1 was used. TABLE 3 presents the M&R and residual NPV costs of the three design alternatives.

The estimated LCC in terms of NPV for the three design alternatives was presented in Figure 4. As shown in the figure, the traffic-based method resulted in the highest NPV, while ERAPave PP delivered the least costly section. The LCC for the traffic-based method is 19% and 21% higher compared to the service life-based and ERAPave PP alternatives, respectively. The higher LCC for the traffic-based method can be attributed to the many M&R interventions required to maintain an optimum service condition. As shown in TABLE 3, the traffic-based alternative requires four mill and replace interventions over the designated economic life of 40 years, whereas the other two methods require only three.

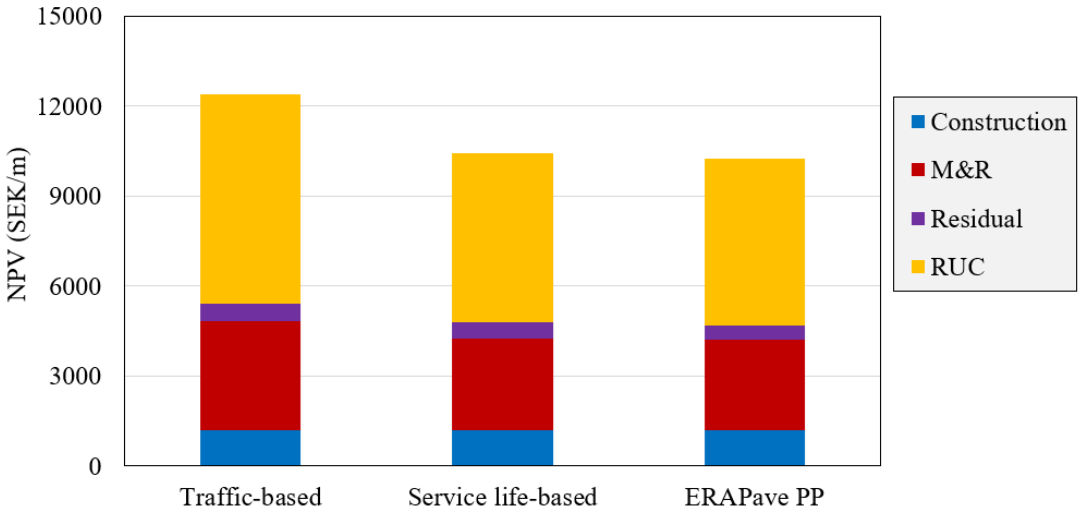


Figure 4 Estimated life cycle cost (LCC) for the three design alternatives

A further analysis of the estimated LCC was performed by closely studying the cost contribution of each cost component. Figure 5 presents the share of each cost component to the estimated LCC for the three design alternatives. The analysis has shown that the contribution of each cost component to the total LCC is approximately the same for the three design alternatives. As shown in Figure 5, RUC comprises almost 60% of the total LCC, making it a significant cost driver. The second important cost item is M&R, which has a share of 30%. The least important cost item is residual cost as its share is limited to a maximum of 5%.

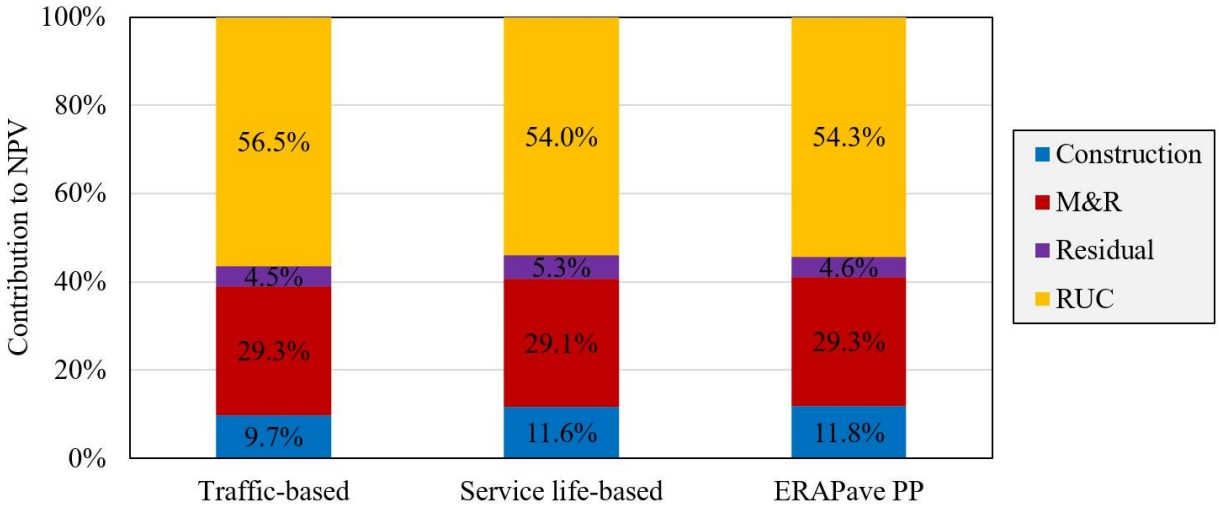


Figure 5 Cost components contribution to the total estimated life cycle cost (LCC)

The average service life of each design alternative was used to establish the relationship between estimated NPVs and long-term performance. Figure 6 presents the relationship between cost components and service life. As shown in the figure, the longer the average service life between treatments, the lower M&R and RU costs will be. The cost for construction remains the same for the three design alternatives as it is independent of future performance. It is important to find M&R actions that result in longer

service life. This is expected to reduce the total number of required M&R treatments, resulting in a lower RUC.

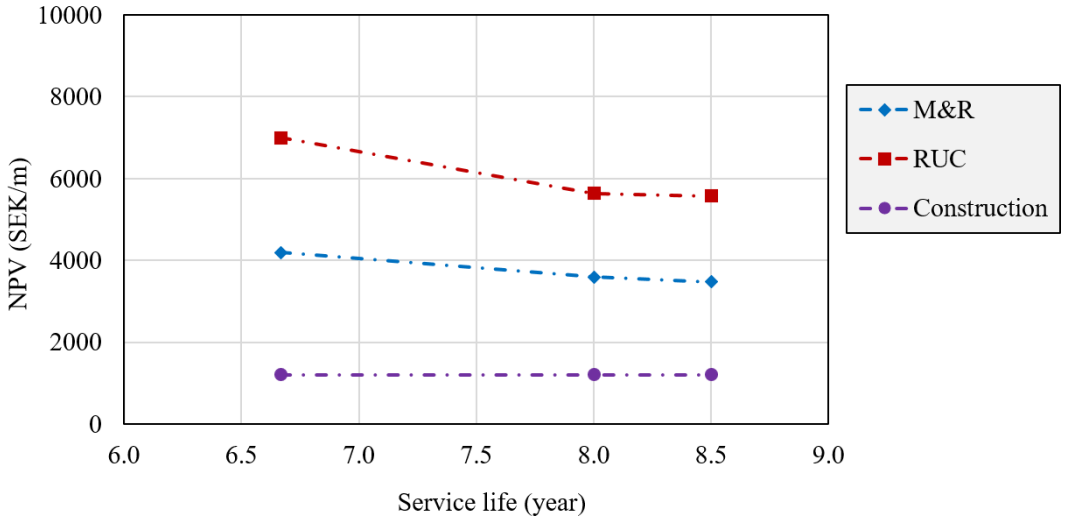


Figure 6 The net present value (NPV) of cost components as defined by service life

The LCCs of the three design alternatives were also calculated using Vägindex as an inflation metric instead of the commonly used CPI. A comparison between the cost components of the two inflation rates for the ERAPave PP design alternative is presented in Figure 7. As shown in the figure, the estimated nominal NPV of a design alternative is highly sensitive to the inflation rate that is chosen. An increase in the inflation rate by 67% is observed to double the total LCC for the ERAPave PP design alternative. The residual cost component is observed to be the most sensitive towards inflation rate fluctuation while construction cost is observed to be the least sensitive. The M&R and RUC components have also increased by 100%.

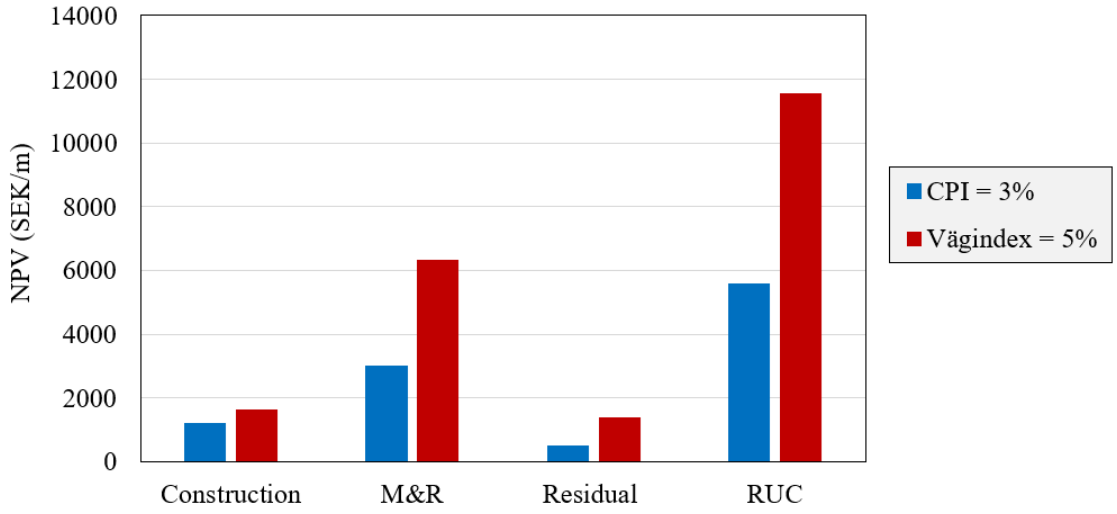


Figure 7 Sensitivity of cost components to changes in inflation rate

In light of the volatility in interest and inflation rates in recent years, a sensitivity analysis was conducted to establish their impact on the estimated LCCs. Various combinations of interest and inflation rates that reflect the volatility in Sweden over the past 25 years were used. The analysis was performed

for the ERAPave PP design alternative using nominal EUAC as an economic indicator. The outcomes of this analysis are presented in Figure 8. As shown in the figure, a combination of high inflation and low interest rates, which was a common phenomenon post-Covid, can significantly increase costs, resulting in budget constraints for road agencies. It has also been reported that the inflation rates of cost components in a road project, such as energy and labor, follow different trends, requiring a comprehensive sensitivity analysis to determine the inflation and interest rates that result in an optimal LCC (*25*).

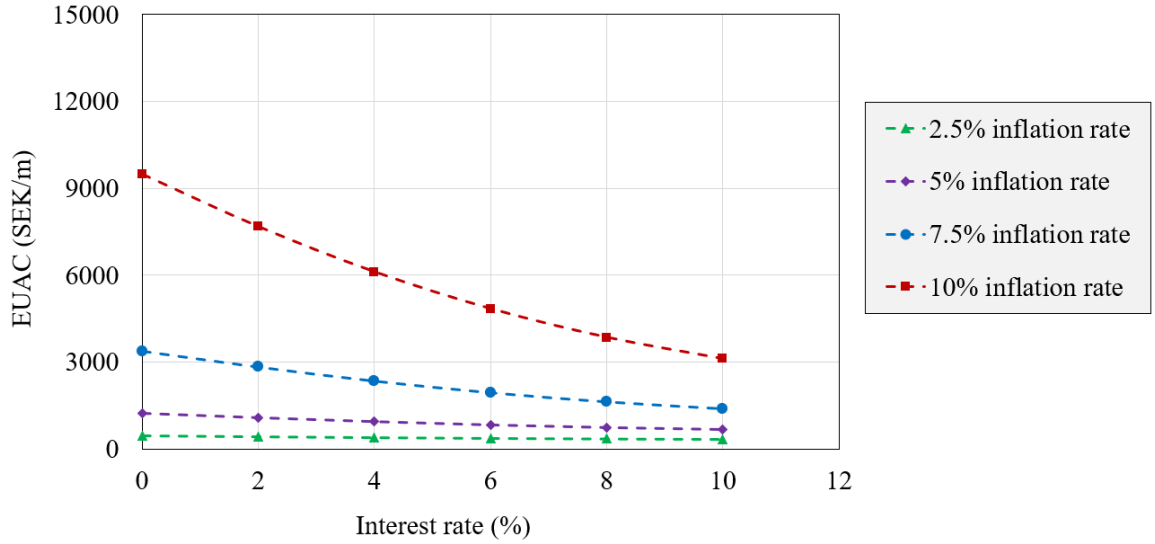


Figure 8 Sensitivity of Equivalent Uniform Annual Cost (EUAC) to interest and inflation rates

CONCLUSIONS

This study evaluates the impact of a pavement design method on predicted long-term performance and, subsequently, on estimated LCC. The analyses were conducted for a collision-free road using three different pavement design methods, as well as various combinations of interest and inflation rates. Based on the obtained results and subsequent discussion, the following conclusions can be drawn:

- Even though the evaluated pavement design methods were developed on competing design philosophies and principles, they have been observed to deliver comparable service lives. This marginal difference in service life between the design alternatives, however, can lead to a significant deviation in estimated LCC, as the impact of minor differences compounds over a long analysis period.
- It has been shown that ERAPave PP delivers the most cost-effective design, while the alternative using the traffic-based method is the most expensive, 21% more costly than the ERAPave PP. This might be due to the fact that, unlike the other two empirical methods, the M-E approach of ERAPave PP provides a better representation of the underlying analysis and design conditions.
- For the analyzed three design approaches, RUC is observed to comprise almost 60% of the total LCC, while the share of M&R is 30%, making these two cost items significant contributors. It can be inferred that designing and constructing a more durable and long-lasting pavement requiring fewer M&R interventions could be a good alternative to reduce LCC and achieve a cost-effective structure.
- The estimated LCCs of the three design alternatives were observed to vary considerably depending on the interest and inflation rates used. This necessitates a comprehensive sensitivity analysis that considers various combinations of interest and inflation rates, facilitating the implementation of a robust risk assessment approach that can identify and minimize unwanted economic constraints.

- The estimated LCC of a given design alternative is highly governed by the utilized unit prices for the various cost categories and cost items. This requires the development of robust and accurate cost modelling approaches that can reliably provide reasonable estimates.

AUTHOR CONTRIBUTIONS

The authors confirm contribution to the paper as follows: study conception and design: Dinegdae and Ahmed; data collection: Dinegdae; analysis and interpretation of results: Dinegdae, Ahmed, Zhu and Erlingsson; draft manuscript preparation: Dinegdae and Erlingsson. All authors reviewed the results and approved the final version of the manuscript.

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DATA ACCESSIBILITY STATEMENT

The datasets generated during and/or analyzed during the current study are available from the corresponding author on reasonable request.

REFERENCES

1. Carlsson, A. *Evaluation of 2+1-Roads with Cable Barrier*. Linköping: Statens väg- och transportforskningsinstitut, p. 49, 2009.
2. Bergh, T., A. Carlsson, and M. Larsson. Swedish Vision Zero Experience. *International Journal of Crashworthiness*, Vol. 8, No. 2, 2003, pp. 159–167. https://doi.org/10.1533/ijcr.2003.0224.
3. Erlingsson, S., S. F. Said, and T. McGarvey. Influence of Heavy Traffic Lateral Wander on Pavement Deterioration. 2012.
4. Wennström, J., H. Sundquist, and R. Karlsson. Life Cycle Cost Considerations in Project Appraisals of Collision-Free Roads. *Structure and Infrastructure Engineering*, Vol. 12, No. 2, 2016, pp. 275–287. https://doi.org/10.1080/15732479.2015.1009125.
5. Wennström, J., and R. Karlsson. Possibilities to Reduce Pavement Rehabilitation Cost of a Collision-Free Road Investment Using an LCCA Design Procedure. *International Journal of Pavement Engineering*, Vol. 17, No. 4, 2016, pp. 331–342. https://doi.org/10.1080/10298436.2014.993191.
6. Lee, E.-B., C. Kim, and J. T. Harvey. Selection of Pavement for Highway Rehabilitation Based on Life-Cycle Cost Analysis: Validation of California Interstate 710 Project, Phase 1. *Transportation research record*, Vol. 2227, No. 1, 2011, pp. 23–32.
7. US Department of Transportation. Economic Analysis Primer. *Management*, 2003, p. 36.
8. Kuttah, D. K. *Advances in Design and Testing of Future Smart Roads: Considering Urbanization, Digitalization, Electrification and Climate Change (1st Ed.).* CRC Press., 2022.
9. Dinegdae, Y., A. Ahmed, S. Rahman, and S. Erlingsson. Evaluation of ERAPave PP Permanent Deformation Models Using APT. *Road Materials and Pavement Design*, Vol. 25, No. sup1, 2023, pp. 95–111. https://doi.org/10.1080/14680629.2023.2191733.
10. Rahman, M. S., S. Erlingsson, and A. Ahmed. Modelling the Permanent Deformation of Unbound Granular Materials in Pavements. *Road Materials and Pavement Design*, 2022. https://doi.org/10.1080/14680629.2022.2108883.
11. Ahmed, A. W., and S. Erlingsson. Numerical Validation of Viscoelastic Responses of a Pavement Structure in a Full-Scale Accelerated Pavement Test. *International Journal of Pavement Engineering*, Vol. 18, No. 1, 2017, pp. 47–59. https://doi.org/10.1080/10298436.2015.1039003.
12. Ahmed, A. W., and S. Erlingsson. Characterisation of Heavy Traffic Axle Load Spectra for Mechanistic-Empirical Pavement Design Applications. *International Journal of Pavement Engineering*, Vol. 16, No. 6, 2015, pp. 488–501. https://doi.org/10.1080/10298436.2014.943131.
13. ISO. *Buildings and Constructed Assets - Service Life Planning Part 5: Life-Cycle Costing.* Geneva, 2008.
14. Walls, J., and M. R. Smith. *Life-Cycle Cost Analysis in Pavement Design: Interim Technical Bulletin.* Washington DC, 1998.
15. AASHTO. *Transportation Asset Management Guide: A Focus on Implementation*. Washington DC:, 2011.
16. Lee, D. B. Fundamentals of Life-Cycle Cost Analysis. *Transportation Research Record*, Vol. 1812, No. 1, 2002, pp. 203–210. https://doi.org/10.3141/1812-25.
17. Levin, P., P. Lilliehorn, and S. Sandesten. *Livscykelekonomi Vid Planering, Byggande Och Förvaltning*. Karslkrona, Sverige, 2008.

18. Ozbay, K., D. Jawad, N. A. Parker, and S. Hussein. Life Cycle Cost Analysis: State-of- the-Practice vs. State-of-the-Art. 2004.
19. Forsman, J. *LCC För Vägbyggnad*. Luleå, 2010.
20. Babashamsi, P., N. I. Md Yusoff, H. Ceylan, N. G. Md Nor, and H. Salarzadeh Jenatabadi. Evaluation of Pavement Life Cycle Cost Analysis: Review and Analysis. *International Journal of Pavement Research and Technology*, Vol. 9, No. 4, 2016, pp. 241–254. https://doi.org/10.1016/j.ijprt.2016.08.004.
21. Karlsson, R., F. Hellman, M. Andersson, and L. G. Wiman. *Restvärde Hos Vägar En Kunskaps- Och Idésammanställning Kring Hur Investeringar*. Linköping:Statens väg- och transportforskningsinstitut, p. 70, 2011.
22. FHWA. *Life-Cycle Cost Analysis Primer*. Washington D.C., 2002.
23. Eisenberger I., Rener D.S, and Lorden G. The Role of Interest and Inflation Rate in Life Cycle Costing Analysis. 42–43.
24. STA. *Impact Analysis of Rehabilitation and Reconstruction of Roads*. Borlänge, 2012.
25. Mirzadeh, I., A. A. Butt, S. Toller, and B. Birgisson. Life Cycle Cost Analysis Based on the Fundamental Cost Contributors for Asphalt Pavements. *Structure and Infrastructure Engineering*, Vol. 10, No. 12, 2014, pp. 1638–1647. https://doi.org/10.1080/15732479.2013.837494.
26. Sund, E. *Life-Cycle Cost Analysis of Road Pavements*. Norges teknisk- naturvitenskapelige universitet, Oslo, 1996.
27. Qin, X., and C. E. Cutler. *Review of Road User Costs and Methods*. Fargo, ND, 2013.
28. Kim, H. B., and S. H. Lee. Reliability-Based Design Model Applied to Mechanistic Empirical Pavement Design. *KSCE Journal of Civil Engineering*, Vol. 6, No. 3, 2002, pp. 263–272. https://doi.org/10.1007/bf02829149.
29. STA. *Underhålls-Standard Belagd Väg [Maintenance Standard Paved Road] In Swedish*. Borlänge, Sweden, 2012.
30. Erlingsson, S., S. Safwat, and T. Mcgarvey. Influence of Heavy Traffic Lateral Wander on Pavement Deterioration. *Proceedings of the 4th European Pavement and Asset Management Conference*, 2012.
31. Said, S. F., H. Hakim, E. Oscarsson, and M. Hjort. Prediction of Flow Rutting in Asphalt Concrete Layers. *International Journal of Pavement Engineering*, Vol. 12, No. 6, 2011, pp. 519–532. https://doi.org/10.1080/10298436.2011.559549.
32. STA. *TRVK Väg: Trafikverkets Tekniska Krav Vägkonstruktion: Anlägningsstyrning. (in Swedish)*. Borlänge, Sweden, 2011.
33. Jacobson, T., and L.-G. Wågberg. *Utveckling Och Uppgradering Av Prognosmodell För Beläggningsslitage Från Dubbade Däck Samt En Kunskapsöversikt Över Inverkande Faktorer. N/-2007 (in Swedish)*. Linköping:, 2007.
34. Ahmed, A. W., and S. Erlingsson. Evaluation of Permanent Deformation Models for Unbound Granular Materials Using Accelerated Pavement Tests. *Road Materials and Pavement Design*, Vol. 14, No. 1, 2013, pp. 178–195. https://doi.org/10.1080/14680629.2012.755936.
35. Barksdale, R. D. Laboratory Evaluation of Rutting in Base Course Materials. 1972.
36. ARA Inc. *Guide for the Mechanistic-Empirical Design of New and Rehabilitated Pavement Structures, Final Report (NCHRP 1-37A)*. Washington, DC, 2004.

37. Rahman, M. S., S. Erlingsson, A. Ahmed, and Y. Dinegdae. A Model for the Permanent Deformation Behavior of the Unbound Layers of Pavements. In *Eleventh International Conference on the Bearing Capacity of Roads, Railways and Airfields, Volume 1* (Hoff, Mork, and Saba, eds.), pp. 277–287.

38. Lundberg, J., S. Janhäll, M. Gustafsson, and S. Erlingsson. Calibration of the Swedish Studded Tyre Abrasion Wear Prediction Model with Implication for the NORTRIP Road Dust Emission Model. *International Journal of Pavement Engineering*, Vol. 22, No. 4, 2021, pp. 432–446. https://doi.org/10.1080/10298436.2019.1614585.

39. Karlsson, R. *Underlag till schabloner för DoU-kostnader i EVA och Vägverkets Effektsamband för nybyggnad och förbättring*. Linköping, 2008.